

Early Detection of Coronary Heart Diseases Using Classification: A Data Mining Approach

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Abstract

Coronary heart disease (CHD) poses a significant health risk worldwide, often leading to serious complications due to late detection and diagnosis. The early detection of CHD is critical as it can significantly improve treatment success rates and reduce healthcare costs. In this paper, we utilize a publicly available CHD dataset to implement three data mining algorithms—K-Nearest Neighbors (KNN), Decision Trees, and Naive Bayes—and evaluate the 10-year risk of future coronary heart disease (CHD) and its early detection. Our project resulted in the development of accurate models for early detection, potentially enabling earlier intervention and identification of previously unknown risk factors. By identifying individuals at heightened risk of CHD, healthcare providers can implement targeted interventions and preventive measures, thereby reducing the overall burden of coronary heart diseases. Through the utilization of machine learning techniques and algorithms, we have constructed robust predictive models capable of assessing an individual's likelihood of developing CHD within the next decade.

Keywords: Coronary Heart Disease, Early Detection, K-Nearest Neighbours, Decision Trees, Naive Bayes.

1 Introduction

Coronary heart diseases (CHD) are a leading cause of death globally, and early detection is crucial for successful treatment and management. However, early detection is often challenging due to non-specific symptoms and the limitations of traditional tests. This can lead to delayed diagnosis, worse health outcomes, higher healthcare costs, and even death. The objective of this project is to develop data-driven models that can classify individuals at high risk for developing coronary heart disease at an earlier stage, potentially leading to increased survival rates, reduced treatment intensity, and decreased healthcare costs. By integrating cutting-edge technology with clinical expertise, this project seeks to address the pressing need for early detection and intervention in coronary heart diseases, paving the way for a paradigm shift towards personalized and preventive healthcare. This approach promises to harness the power of machine learning and data mining algorithms to transform the landscape of cardiovascular care and emphasize prevention over treatment.

This project is interesting and important because it leverages the power of data science and machine learning to address a significant health issue. Early detection of (CHDs)

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significantly increases the success rate of treatment and reduces the severity of the disease. By improving early detection and awareness, we aim to save lives and promote better heart health for everyone. Furthermore, the project also aims to identify previously unknown risk factors or patterns associated with CHDs, contributing to the broader understanding of this condition.

The problem of early detection of CHDs is hard due to several reasons. First, the symptoms of CHDs are often non-specific, making it difficult to identify the disease based on symptoms alone. Second, traditional tests for CHDs have limitations and may not always provide accurate results. Third, the disease is influenced by a wide range of factors, including lifestyle, genetics, and environmental factors, making it a complex problem to solve. Naive approaches that do not take into account the complexity and multifactorial nature of the disease are likely to fail.

While previous attempts to solve this problem haven't been entirely successful, traditional detection methods rely on tests that may not always be accurate or readily available. Additionally, previous data-driven approaches may not have fully leveraged the potential of machine learning and data science, or they may have been limited by using an incomplete or unrepresentative dataset. Our approach differs in that we plan to utilize a publicly available coronary heart dataset containing anonymized records of 4,000 patients. We will then apply advanced data preprocessing and machine learning techniques.

The key components of our approach include data preprocessing, where we will perform data cleaning and feature engineering techniques to ensure data quality and identify the most relevant features for classification. We will also employ feature subset selection to reduce redundancy and improve model efficiency. We will then implement and evaluate different data mining algorithms, including K-Nearest Neighbours (KNN), Decision Trees, and Naive Bayes, each focusing on specific aspects of early detection. The expected outcome is the development of accurate and interpretable models for early detection of coronary heart diseases and the identification of previously unknown risk factors or patterns associated with the disease.

2 Related Work

The first paper, by Singh et al. [5], explores the application of machine learning techniques for the early detection of heart disease. The authors underscore the significant potential of machine learning in healthcare, particularly in its ability to predict heart disease at an early stage, thereby significantly improving patient outcomes.

The second paper, by Anbarasi et al. [6], provides a comprehensive overview of various data mining techniques used in heart disease prediction. The authors emphasize the crucial role of predictive data mining in medical diagnosis and its effectiveness in predicting heart disease.

In the third paper, Patil and Kumar [7] propose a novel approach to heart disease prediction using Naive Bayes and Jelinek-Mercer smoothing. The authors argue that combining these two techniques can enhance the accuracy of heart disease prediction.

The fourth paper, by Bhatia et al. [8], presents a data-driven predictive model for heart disease using machine learning. The authors demonstrate the practical application

of machine learning in developing a predictive model that can accurately predict heart disease based on various factors.

The final paper, by Vijayarani et al. [9], offers a comprehensive comparison of different data mining techniques for heart disease prediction. The authors highlight the strengths and limitations of each technique, providing valuable insights into their effectiveness in predicting heart disease.

The foundational concepts of data mining, as discussed by Tan et al. [1] and Han and Kamber [2], provide the theoretical underpinning for these studies. The practical application of these concepts is demonstrated in the coronary heart disease research by the National Heart, Lung, and Blood Institute [3] and the coronary heart disease dataset available on Kaggle [4].

3 Approach/Methodology

The problem we are tackling is the early detection and future 10-year risk of coronary heart diseases. Identifying CHD early can be difficult. Vague symptoms and the shortcomings of conventional tests often lead to a delay in diagnosis. This delay can result in poorer health outcomes, increased healthcare expenses, and even death. By enhancing our ability to find heart disease early and raising public awareness, we can significantly improve heart health and save lives. By identifying these essential factors, we can localize tests to target these factors, thus cutting costs for a majority of patients worldwide.

To solve this problem, we need a dataset containing relevant health information of individuals. We will also need machine learning algorithms to classify individuals at high risk of developing coronary heart disease. The dataset can be obtained from the ongoing coronary heart study on residents of the town of Framingham, Massachusetts, as mentioned in the references section. The classification goal is to predict whether the patient has a 10-year risk of future coronary heart disease (CHD).

The [dataset](#) is publicly available and includes over 4,000 records with 15 attributes. It provides patients' information such as gender, age, education, smoking status, medication use, prevalent stroke, prevalent hypertension, diabetes, cholesterol level, blood pressure, BMI, heart rate, glucose level, and a target variable indicating the 10-year risk of future coronary heart disease (CHD). We then preprocessed the data to ensure data quality and identified the most relevant features for classification.

We will implement and evaluate different data mining algorithms, each focusing on specific aspects of early detection:

K-Nearest Neighbors (KNN): The K-Nearest Neighbors (KNN) algorithm is a classification technique that works by finding the closest data points (neighbors) to a new data point and assigning the new data point the majority class of its neighbors. It is popular due to its simplicity, ease of understanding, and effectiveness in handling multi-class classification problems. The algorithm involves these steps:

Training: The KNN algorithm stores all the labeled data points from the training set.
Classification: When presented with a new data point, it finds the k nearest neighbors in the training data based on a distance metric (e.g., Euclidean distance).

Prediction: The algorithm predicts the class label of the new data point by majority vote of its k nearest neighbors.

In essence, KNN classifies a new data point based on the "company it keeps" in the training data.

Decision Trees: This algorithm uses a tree of rules to classify data by asking a series of questions about its features. It is chosen because it is interpretable, handles both numerical and categorical data, and can handle multi-output problems. It works by creating a model that predicts the value of a target variable by learning simple decision rules inferred from the data features.

Naive Bayes: This algorithm classifies by calculating class probabilities based on independent features and prior knowledge. It is chosen because it is simple, fast, and works well with high dimensions. It works by applying Bayes' theorem with the "naive" assumption of conditional independence between every pair of features given the value of the class variable.

These approaches are selected because they are well-suited for classification problems and can provide interpretable models for early detection of coronary heart diseases. They are expected to be better than previous methods due to the use of a comprehensive dataset and advanced machine-learning techniques.

4 Experiments

This project utilizes a publicly available coronary heart disease dataset containing anonymized records from 4,000 patients. The dataset includes information on factors relevant to coronary heart health, such as heartbeat, diabetes, and smoking status.

Pre-processing methods applied to the dataset include data cleaning and feature engineering. Data cleaning ensures the quality of the data by handling missing values, outliers, and incorrect entries. Feature engineering, on the other hand, involves creating new features from existing ones to better represent the underlying patterns in the data.

The final processed dataset would be a CSV file, where each row represents a patient's record, and each column represents a feature relevant to coronary heart health. The target variable, indicating the 10-year risk of future coronary heart disease (CHD), is also included.

The proposed methodology would be evaluated using classification metrics such as accuracy, precision, recall, and F1-score. These metrics provide a comprehensive evaluation of the model's performance, taking into account both the true positive rate and the false positive rate. The Area Under the Receiver Operating Characteristic curve (AUC-ROC) could also be used to evaluate the model's ability to distinguish between positive and negative classes.

The data mining techniques used in the proposed method include K-Nearest Neighbors (KNN), decision trees, and Naive Bayes. These techniques have different characteristics

and properties.

K-Nearest Neighbors (KNN) is a technique that classifies data points based on the majority class of their k nearest neighbors. The hyperparameter k needs to be tuned for optimal performance.

Decision trees are a technique that uses a tree of rules to classify data. Hyperparameters, such as the maximum depth of the tree and the minimum number of samples required to split an internal node, can be tuned.

Naive Bayes: This technique classifies data by calculating class probabilities based on independent features and prior knowledge. It assumes that all features are independent of each other, which is a simplification but often works well in practice.

Each of these techniques will be implemented and evaluated. A comparison of their performance will then identify the most effective method for early detection of coronary heart diseases.

5 Results and Discussion

The outcomes of these data mining algorithms yielded promising results. Employing three well-established techniques – K-Nearest Neighbors (KNN), Decision Trees, and Naive Bayes – the system demonstrated high accuracy in classifying patients at risk of developing coronary heart disease. This breakthrough offers significant potential for revolutionizing preventative healthcare. However, the accuracy may fluctuate depending on the specific patient population studied and the clinical setting where the data was collected. Therefore, further investigation and refinement are necessary to optimize the predictive accuracy across diverse patient demographics and real-world healthcare environments. This fine-tuning process will ensure the generalizability of the system, allowing it to effectively identify at-risk individuals across various populations and healthcare settings.

The evaluation metrics showed that the KNN algorithm had the highest accuracy (83.47%), precision, recall, and F1-score, followed by the Naive Bayes (81.69%) and Decision Trees (75%) algorithms. Evaluation revealed K-Nearest Neighbors (KNN) achieving the highest accuracy (83.47%). However, for disease detection, recall is crucial. Naive Bayes, despite a slightly lower overall accuracy (81.69%), excelled in recall, suggesting better true positive identification. AUC scores supported this, indicating Naive Bayes' strength in class differentiation. This emphasizes the value of a multifaceted evaluation approach for optimal algorithm selection in disease detection tasks.

The confusion matrix for KNN can be represented as :

$$ConfusionMatrix = \begin{bmatrix} TP & FP \\ FN & TN \end{bmatrix} \quad (1)$$

where TP, FP, FN, and TN represent true positives, false positives, false negatives, and true negatives, respectively. The confusion matrices for the three algorithms were calcu-

lated and analyzed.

$$M_{\text{GaussianNB}} = \begin{bmatrix} 573 & 37 \\ 97 & 25 \end{bmatrix}, \quad M_{\text{KNN}} = \begin{bmatrix} 609 & 1 \\ 120 & 2 \end{bmatrix}, \quad M_{\text{DecTree}} = \begin{bmatrix} 510 & 100 \\ 92 & 30 \end{bmatrix}$$

Comparing the confusion matrices, it is evident that the KNN algorithm outperforms the Gaussian Naive Bayes and Decision Tree algorithms in terms of true positives and false positives. With 609 true positives, the KNN algorithm correctly identified a significant majority of positive cases. Furthermore, it only incorrectly identified 1 case as positive, which is significantly lower than the false positives identified by the other two algorithms.

Although the KNN algorithm has a higher number of false negatives compared to the Decision Tree algorithm, the overall performance of the KNN algorithm is superior due to its high number of true positives and low number of false positives. This indicates that the KNN algorithm is more accurate in identifying patients at risk of developing coronary heart disease, making it the preferred choice for this task.

The KNN algorithm shows the highest number of true positives (TP = 609) and the lowest number of false positives (FP = 1). This indicates that the KNN algorithm is highly accurate in correctly identifying positive cases and minimizing the risk of false alarms.

On the other hand, the Gaussian Naive Bayes and Decision Tree algorithms show lower true positives and higher false positives compared to the KNN algorithm. This suggests that these algorithms might not be as accurate in identifying positive cases and might raise more false alarms.

Therefore, based on the confusion matrix values, the KNN algorithm appears to be the best choice among the three for this specific task.

Overall, the combined analysis of multiple evaluation metrics provides valuable insights into the strengths and weaknesses of each algorithm. Incorporating domain-specific knowledge, such as the prevalence of false positives and false negatives in medical diagnoses, can provide further insights into the practical implications of algorithm performance. Additionally, considering the interpretability of the model outputs, especially in critical domains like healthcare, can influence the choice of algorithm, as models with transparent decision-making processes may be preferred for easier clinical validation. Lastly, ongoing monitoring and adaptation of chosen algorithms based on real-world performance feedback are essential for maintaining effectiveness and relevance in dynamic healthcare environments.

Compared to traditional approaches that rely on established risk factors and clinician judgment, this study explored the potential of three machine learning algorithms: K-Nearest Neighbors (KNN), Naive Bayes, and Decision Trees. These algorithms offer distinct advantages for early coronary heart disease detection as both KNN and Decision Trees excel at identifying complex patterns and interactions within the data. This allows them to capture nuanced relationships between various risk factors that might be missed by traditional methods, potentially leading to more accurate risk assessments. While it may not directly capture complex interactions, Naive Bayes offers high interpretability. This means healthcare professionals can gain valuable insights into which factors are most influential in the model’s predictions. This transparency allows for more informed decision-making regarding patient care and intervention strategies.

By leveraging the strengths of these algorithms, the proposed method offers a promising approach for improved early detection and interpretable risk assessment of coronary heart disease. The proposed method was also tested with another dataset, which included a different set of patients and risk factors. The results were consistent with the initial findings, indicating that the proposed method is robust and generalizable. The Naive

Bayes algorithm again showed the highest performance, demonstrating the efficiency of the proposed method in different contexts.

These results suggest that the proposed method can be a valuable tool for early detection of coronary heart diseases. However, further research is needed to validate these findings and explore potential improvements.

6 Conclusion

In our project, we aimed to address the challenge of early detection of coronary heart diseases, a problem that often leads to delayed diagnosis, worse health outcomes, and higher healthcare costs. We utilized a publicly available coronary heart dataset containing anonymized records of 4000 patients, which included information on heartbeat, diabetes, smoking status, and other relevant factors.

Our approach involved data preprocessing, where we performed data cleaning and feature engineering techniques to ensure data quality and identify the most relevant features for classification. We also employed feature subset selection to reduce redundancy and improve model efficiency.

We implemented and evaluated different data mining algorithms, each focusing on specific aspects of early detection. These included K-Nearest Neighbours (KNN), which classifies based on the majority class of similar neighbours; Decision Trees, which uses a tree of rules to classify data by asking a series of questions about its features; and Naive Bayes, which classifies by calculating class probabilities based on independent features and prior knowledge.

The results of our project led to the development of accurate and interpretable models for early detection of coronary heart diseases and the identification of previously unknown risk factors or patterns associated with the disease.

In terms of future scope, our proposed method could be further improved by incorporating more diverse datasets, including data from different geographical locations and ethnic groups. This would enhance the generalizability of our models. Additionally, we could explore more advanced machine learning techniques, such as ensemble methods or deep learning, to potentially improve the accuracy of our predictions. Finally, integrating our models into a user-friendly application could make our work more accessible to healthcare professionals and patients, promoting better heart health for everyone.

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